Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)			ΔU: is the increase in the internal energy of a system (1) accept change Q: is the heat entering (1) the system W: is the work done <u>by</u> (1) the system "system" or "gas" or equivalent needs to be used once correctly for full marks.	3			3		
	(b)			Substitution into $PV = nRT$ (1) $T = \frac{PV}{nR} = \frac{(100\times10^3)(1.4\times10^{-3})}{(0.06)((8.31)}$ = 280.8 K **unit mark** (1)	1	1		2	2	
	(c)	(i)		Work done (by gas) = $P\,\Delta V = (100 \times 10^3)(2.0 - 1.4) \times 10^{-3}(1)$ $= 60$ [J] (1) Award 1 mark for 200 or 140 [J]	1	1		2	2	
		(ii)		Final temperature: $T = \frac{PV}{nR} = \frac{(100\times10^3)(2.0\times10^{-3})}{(0.06)((8.31)} = 401.1$ [K] (1) Increase in internal energy = $\frac{3}{2}nR\,\Delta T =$ $1.5(0.06)(8.31)(401.1 - 280.8\textbf{ecf}) = 90.0$ [J] (1) **Alternative**: Use of $\frac{3}{2}P\,\Delta V$ (1) = 90 [J] (1)		2		2	2	
		(iii)		Heat transferred $Q = \Delta U + W = 90.0\ \textbf{ecf} + 60.0\ \textbf{ecf} = 150.0$ [J]		1		1	1	
				Question 1 total	**5**	**5**	**0**	**10**	**7**	**0**